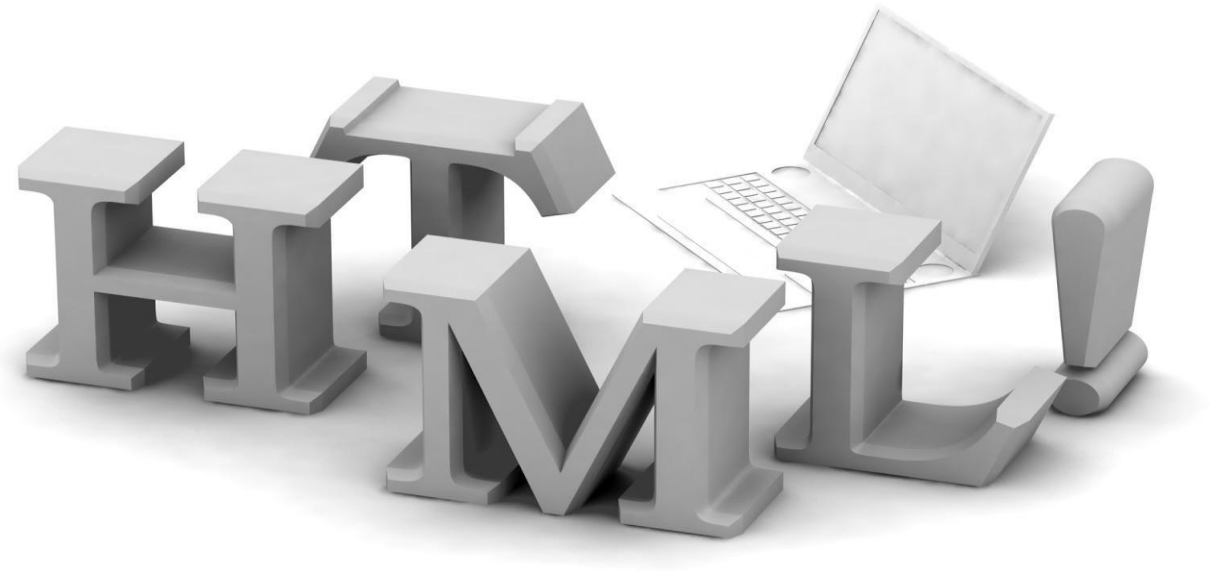
Faculty of Computer Science and Mathematics

Lab Exercise 4

Name:	**Fasya Nasihah binti Razali**
Matric Num:	**S79235**



Overview:

This in-class exercise extends Lab Exercise 4. You will apply HTML5 page structure elements, embed a video using the <video> tag, and use CSS (internal and external) to style a web page. You will practise CSS syntax, ID and class selectors, a CSS navigation bar, text decoration and link states (:link, :visited, :hover, :active).

There are two problems. Your lecturer will assign one problem to you, or you may be asked to complete both.

Learning Outcomes

1. Utilize semantic HTML5 elements to create a well-structured web page layout.
2. Embed video content using the <video> and <source> tags.
3. Apply CSS syntax using internal and external style sheets, with ID and class selectors.
4. Build a functional and visually appealing navigation bar using CSS.
5. Use text-decoration and link states to improve text presentation and user experience.

Best of luck!

Problem A: Visit Terengganu – Tourism Promotion Page

The state tourism office wants a simple promotional web page to attract students and visitors to Terengganu. The page must look attractive, have a working navigation bar, show a promotional video, and include a trip interest form. Build the page using HTML5 and CSS (both internal and external CSS). You may choose any destinations, prices and video.

Instructions

Create two files named lab4_ExerA.html and style.css that contain the following:

1. HTML5 Page Structure

- Use `<!DOCTYPE html>`, `<meta charset="utf-8">` and `<meta name="viewport" content="width=device-width, initial-scale=1.0">`.
- Structure the page using `<header>`, `<nav>`, at least three `<section>` elements with ids (e.g. destinations, video, trip) and a `<footer>`.
- In the Destinations section, use at least three `<article>` elements. Each article has an `<h3>` (place name), a short `<p>` description and a promotion price.
- Add a `<meter>` element showing how many trip seats have been filled (e.g. min="0" max="40" value="28").
- In the footer, add © and an `<address>` element with an e-mail link (mailto:).

2. Navigation Bar

- Create the navigation bar using an unordered list (`<ul>`, `<li>`, `<a>`) with links to each section (e.g. href="#video").
- In style.css, remove the bullets (list-style-type: none), remove the underline from the links (text-decoration: none), and set background-color, padding, text-align and texttransform: uppercase.
- Change the background colour of a menu item when the mouse is over it (:hover).

3. Insert Media (Video)

- Embed a video in the Video section using `<video>` with width, height and controls, a `<source>` with the correct type (e.g. video/mp4), and fallback text for browsers that do not support video.
- Add a caption below the video (e.g. using `<figure>` and `<figcaption>`, or a `<p class="caption">`).

4. CSS Styling (Internal and External)

- Link style.css using `<link rel="stylesheet" href="style.css">`. Most of your styles must be in this file.
- Add a `<style>` block in the `<head>` (internal CSS) with at least two rules for this page only.

- Use at least two ID selectors (e.g. #destinations) and at least two class selectors (e.g. .card, .price, .center).
- Use at least six different CSS properties, e.g. color, background-color, font-family, fontsize, text-align, padding, border, width.

5. Text Decoration

- Apply text-decoration to headings, e.g. overline for <h1> and underline for <h2>.
- Show each promotion with an old price and a new price. Style the old price using a class with text-decoration: line-through (e.g. RM 299 crossed out, followed by RM 199).

6. Interactive Link States

- Style a:link, a:visited, a:hover and a:active with different colours. Write them in the correct order (link → visited → hover → active) so that all states work.

7. Interactive Elements (Form)

- In the Trip section, include a form with the following fields: – Name (text input) and E-mail (type="email")
  - Traveller type (radio buttons): Student, Staff, Public
  - Interests (checkboxes): Islands, Food, Culture, Nature – Submit button
- Style the form using a class (e.g. .trip-form) with background-color, padding, border and width.

8. Responsive Design and Comments

- Make the page fit small screens: use percentage widths (e.g. width: 90%) and make the video scale with video { max-width: 100%; height: auto; }. Test by resizing the browser window.
- Use comments to explain each section of your HTML (<!-- ... -->) and CSS (/* ... */) code.

Submission

Save the files as lab4_ExerA.html and style.css, validate the HTML at https://validator.w3.org and the CSS at https://jigsaw.w3.org/css-validator (Validate by File Upload), and submit them (Submit lab4_ExerA.html, style.css and Screenshot of validation) through Github > Update your index.html (if you are hosting your lab in Github pages)

Validation :

Nu Html Checker

his tool is an ongoing experiment in better HTML checking, and its behavior remains subject change

howing results for uploaded file lab4_ExerA.html (checked with vnu 26.11.10)

Checker Input

Show ☐ source ☐ outline ☐ image report ☐ errors & warnings only Options…

Check by file upload Choose File No file chosen

Uploaded files with .xhtml or .xht extensions are parsed using the XML parser.

Check

ocument checking completed. No errors or warnings to show.

ed the HTML parser.

tal execution time 5 milliseconds.

out this checker • Report an issue

W3C CSS Validator results for style.css (CSS level 3 + SVG)

Congratulations! No Error Found.

This document validates as CSS level 3 + SVG !

To show your readers that you've taken the care to create an interoperable Web page, you may display this icon on any page that validates. Here is the XHTML you could use to add this icon to your Web page:

```
<p>
    <a href="https://jigsaw.w3.org/css-validator/check/referer">
        <img style="border:0;width:88px;height:31px"
            src="https://jigsaw.w3.org/css-validator/images/vcss"
            alt="Valid CSS!" />
    </a>
</p>
```

```
<p>
    <a href="https://jigsaw.w3.org/css-validator/check/referer">
        <img style="border:0;width:88px;height:31px"
            src="https://jigsaw.w3.org/css-validator/images/vcss-blue"
            alt="Valid CSS!" />
    </a>
</p>
```

Output:

Discover Terengganu

Beautiful Islands, Culture and Nature

HOME DESTINATIONS VIDEO PLAN YOUR TRIP

Welcome to Terengganu

Discover the beauty of Terengganu, a beautiful state in Malaysia famous for its islands, beaches, traditional culture and delicious local food. Enjoy an unforgettable holiday experience with your family and friends.

Popular Destinations

Redang Island

Redang Island is famous for its crystal-clear water, beautiful beaches and amazing marine life.

~~RM 200~~ **RM 199**

Perhentian Islands

Enjoy snorkelling, diving and relaxing on the beautiful beaches of Perhentian Islands.

~~RM 350~~ **RM 250**

Kapas Island

Kapas Island is famous for its white sandy beaches, crystal-clear water and peaceful tropical atmosphere. Visitors can enjoy swimming, snorkelling and relaxing.

~~RM 180~~ **RM 120**

Trip seats filled: 28 out of 40

Explore Terengganu





Discover the beautiful attractions of Terengganu.

Plan Your Trip

Fill in the form below to share your travel interests.

Name:

E-mail:

Traveller Type:
○ Student ○ Staff ○ Public

Interests:
☐ Islands ☐ Food ☐ Culture ☐ Nature

Submit

Problem B: e-Café Campus (Debugging and Extend)

The campus café has a promotional page, but it does not look as expected: the menu still shows bullet points, the heading has no line above it, the prices are not highlighted, the hover colour does not work on visited links, the "Menu" link does nothing and the video cannot be played. Find and fix the errors, then extend the page.

Given Code

```
<!DOCTYPE html>
<html lang="en">
<head>
<meta charset="utf-8">
<title>e-Café Campus</title>
<style type="text/css">
<!-- Navigation bar -->
ul {
list-style-type: no;
margin: 0;
padding: 0;
}
a:link {color: #1A5276;}
a:hover {color: #E67E22;}
a:visited {color: #7D3C98;}
a:active {color: #C0392B;}
h1 {text-decoration: overlined;}
p {color #333333; text-align: justify;}
#intro {
font-size: 18px;
background-color: #FDEBD0;
highlight {
font-weight: bold;
color: #C0392B;
}
</style>
</head>
<body>
<header>
<h1>e-Café Campus</h1>
</header>
<nav>
<ul>
<li><a href="menu">Menu</a></li>
<li><a href="#video">Video</a></li>
<li><a href="#contact">Contact</a></li>
</ul>
</nav>
<section id="menu">
<h2>Today's Menu</h2>
<p class="intro">Freshly brewed coffee and local delights every
day.</p>
<article id="item">
<h3>Kopi Tarik</h3>
```

```
44          <p>Only <span class="highlight">RM 3.50</span></p>
45          </article>
46          <article id="item">
47          <h3>Nasi Lemak</h3>
48          <p>Only <span class="highlight">RM 5.00</span></p>
49          </article>
50          </section>
51          <section id="video">
52          <h2>Watch Our Story</h2>
53          <video width="400" height="300">
54          <source src="cafe.mp4" type="mp4">
55          Your browser does not support the video tag.
56          </video>
57          </section>
58          <footer id="contact">
59          <p>&copy; 2026 e-Café Campus</p>
60          </footer>
61          </body>
62          </html>
```

Instructions

Part 1. Find and Fix the Errors

1. Copy the given code into a new file named lab4_ExerB.html.

2. Identify and correct all errors (at least 8). At the top of the <body>, write an HTML comment listing each error and the line number, e.g. <!-- Error 1 (Line 9): ... -->.

3. Test the page: the menu has no bullet points, the <h1> has a line above it, the prices are bold and red, the "Menu" link jumps to the menu section, the hover colour works on all links, and the video can be played.

4. Validate your corrected file at https://validator.w3.org and your CSS at https://jigsaw.w3.org/css-validator. Your final page must show no errors.

Part 2. Extend the Existing Page

5. Create an external style sheet named cafe.css and link it to the page. Move the navigation bar styles into this file, then style the navigation bar: links displayed as blocks , text-decoration: none, and a different background colour on :hover.

6. Add a promotion to each menu item: show the old price using a class .old-price with text-decoration: line-through, followed by the new price (e.g. RM 4.50 crossed out, followed by RM 3.50).

7. Add a caption below the video using `<figure>` and `<figcaption>`. Add the viewport `<meta>` tag and make the video responsive with max-width: 100% and height: auto.

8. Add a new `<section id="feedback">` with a form containing: Name, E-mail, Visit type (radio: Dine-in, Take-away), Favourite items (at least 3 checkboxes) and a Submit button. Style the form using a class `.feedback-form`, and add a link to this section in the navigation bar.

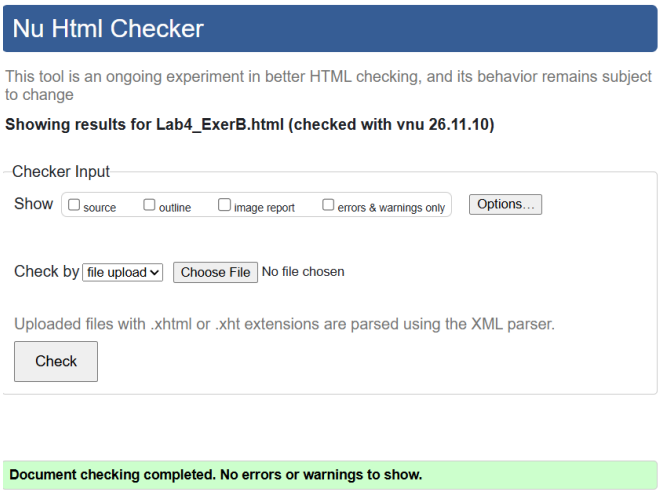
9. In the footer, add an `<address>` element with an e-mail link (`mailto:`).

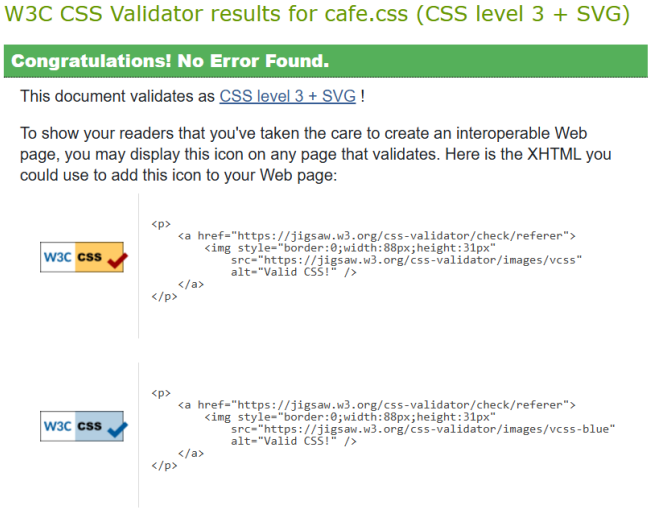
10. Use comments to explain each new section of your HTML and CSS code.

Submission

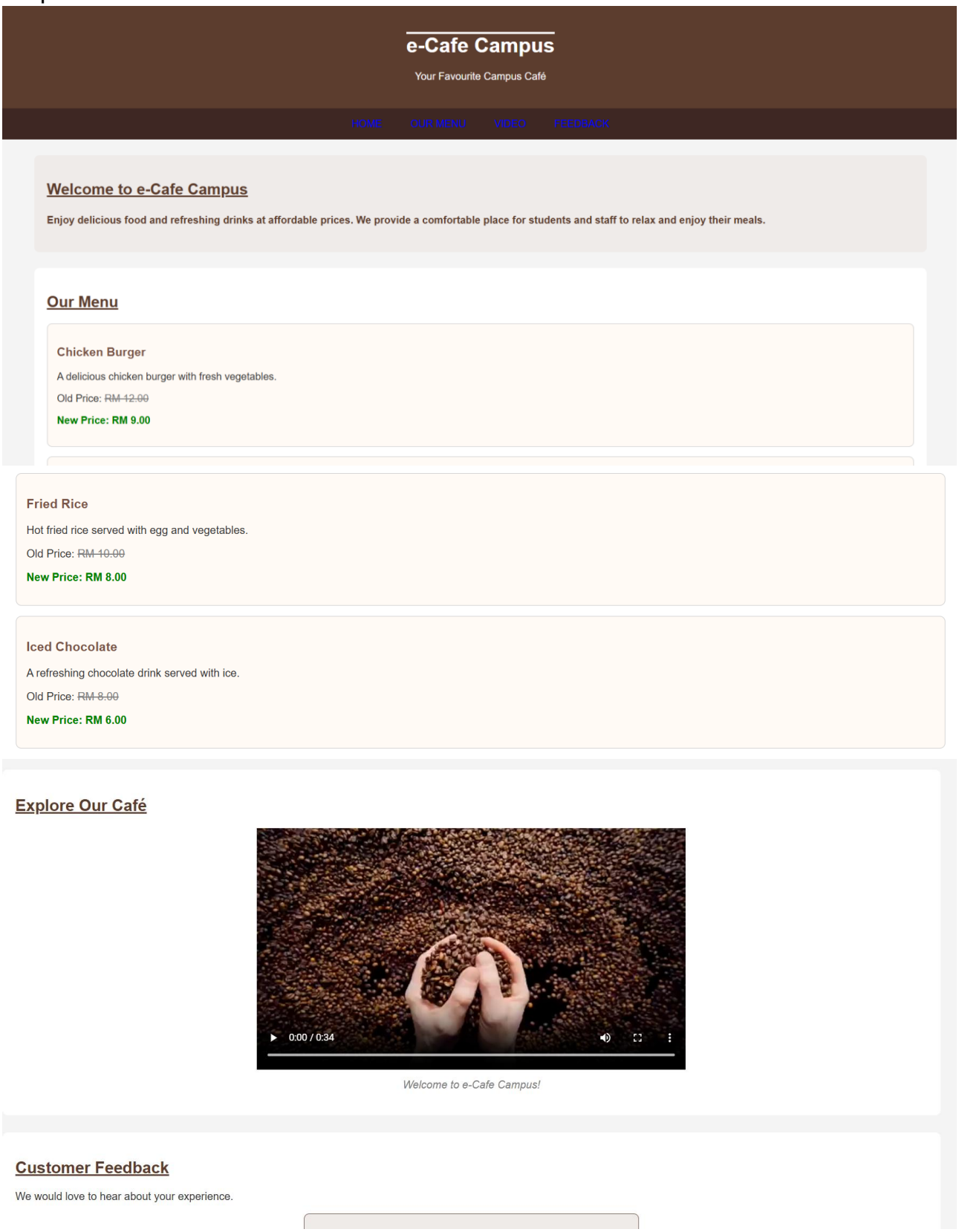
Save the files as lab4_ExerB.html and cafe.css, validate them at https://validator.w3.org and https://jigsaw.w3.org/css-validator (Validate by File Upload), and submit them (Submit lab4_ExerB.html, cafe.css and Screenshot of validation) through Github

Validation





Output:


e-Cafe Campus
Your Favourite Campus Café
HOME
OUR MENU
VIDEO
FEEDBACK
Welcome to e-Cafe Campus
Enjoy delicious food and refreshing drinks at affordable prices. We provide a comfortable place for students and staff to relax and enjoy their meals.
Our Menu
Chicken Burger
A delicious chicken burger with fresh vegetables.
Old Price: RM 12.00
New Price: RM 9.00
Fried Rice
Hot fried rice served with egg and vegetables.
Old Price: RM 10.00
New Price: RM 8.00
Iced Chocolate
A refreshing chocolate drink served with ice.
Old Price: RM 8.00
New Price: RM 6.00
Explore Our Café
0:00 / 0:34
Welcome to e-Cafe Campus!
Customer Feedback
We would love to hear about your experience.

Customer Feedback

We would love to hear about your experience.

Name:

E-mail:

Type of Visit:
○ Dine-in ○ Take-away

Favourite Items:
☐ Burger ☐ Fried Rice ☐ Drinks

Submit Feedback

© 2026 e-Cafe Campus. All Rights Reserved.

Contact us: ecafe@example.com

Lab 4 Exercise

4.9 Lab Exercise 4

In this exercise, you will build a functional webpage that integrates various HTML5 and CSS concepts. You will apply semantic HTML5 elements to structure content effectively, incorporate embedded media for enhanced interactivity, and use CSS to create a visually appealing design.

This lab aims to reinforce your understanding of modern web development principles while improving your ability to design professional and user-friendly webpages. By the end of this task, you will have a well-structured and styled webpage that demonstrates your ability to combine HTML5 and CSS effectively.

4.9.1 Building a Responsive Webpage with HTML5 and CSS

Instruction

1. **Create the HTML File:**
   - Name the file **lab4.html**.
2. **Define the Structure:**
   - Use semantic HTML5 elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` to structure your webpage.
     - Create a navigation bar using an unordered list styled with CSS. Include links to various sections of the page.
3. **Insert Media**
   - Embed a video in one section of the webpage.
   - Ensure the video is properly captioned and fits the design of your page.
4. **Apply CSS Styling**
   - Use both internal and external CSS to style the page.
   - Apply text decoration to headings and links using CSS properties such as `text-decoration`: underline, overline, line-through, and others.
   - Create a responsive design by styling your links for various states (`:hover`, `:visited`, `:active`).
5. **Interactive Elements**
   - Include a form with the following fields:
     - Name (Text Input)
     - Email (Text Input)
     - Gender (Radio Buttons)
     - Favorite Hobbies (Checkboxes)
     - Submit Button
   - Style the form using CSS.

Submission Requirements

- A single HTML file linking to an external CSS file.
- The webpage must follow good design practices and be fully functional in modern browsers.

Coding : Lab4.html

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">

    <title>My Personal Profile</title>

    <!-- External CSS -->
    <link rel="stylesheet" href="style.css">

    <!-- Internal CSS -->
    <style>
        .welcome-text {
            color: #ffffff;
            font-size: 18px;
            text-align: center;
            margin-top: 10px;
        }

        .highlight {
            color: #f4c27a;
            text-decoration: underline;
        }
    </style>
</head>

<body>

    <!-- Header -->
    <header id="home">
        <h1>Welcome to My Website</h1>
        <p class="welcome-text">
            Discover more about <span class="highlight">My Personal Profile</span>
        </p>
    </header>

    <!-- Navigation Bar -->
    <nav>
        <ul>
            <li><a href="#home">Home</a></li>
            <li><a href="#about">About Me</a></li>
            <li><a href="#video">My Video</a></li>
            <li><a href="#hobbies">Hobbies</a></li>
            <li><a href="#contact">Contact</a></li>
        </ul>
    </nav>

    <!-- Main Content -->
    <main>

        <!-- About Section -->
        <section id="about">
            <h2>About Me</h2>

            <article>
                <h3>Introduction</h3>

                <p>
                    Hello! My name is Fasya. I am a student who
                    enjoys learning new things and exploring
                    different experiences.

                    This website is created to introduce myself,
                    share my interests and hobbies, and provide
                    a little information about me.
                </p>
            </article>
        </section>

        <!-- Video Section -->
        <section id="video">
            <h2>My Favorite Video</h2>

            <figure>
                <video controls width="100%">
                    <source
                        src="https://interactive-examples.mdn.mozilla.net/media/cc0-videos/flower.mp4"
                        type="video/mp4">
```

```
                    Your browser does not support HTML5 video.
                </video>

                <figcaption>
                    Figure 1: A beautiful flower video.
                </figcaption>
            </figure>
        </section>

        <!-- Hobbies Section -->
        <section id="hobbies">
            <h2>My Hobbies</h2>

            <article>
                <p>
                    Everyone has their own interests and hobbies.
                    Here are some of my favorite activities:
                </p>

                <ul>
                    <li>Listening to music</li>
                    <li>Watching kdrama</li>
                    <li>Cafe Hunting</li>
                    <li>Travelling</li>
                </ul>
            </article>
        </section>

        <!-- Contact Form -->
        <section id="contact">
            <h2>Contact Me</h2>

            <form action="#" method="post">

                <!-- Name -->
                <label for="name">Name:</label>
                <input
                    type="text"
                    id="name"
                    id="name"
                    name="name"
                    placeholder="Enter your name"
                    required>

                <!-- Email -->
                <label for="email">Email:</label>
                <input
                    type="email"
                    id="email"
                    name="email"
                    placeholder="Enter your email"
                    required>

                <!-- Gender -->
                <label>Gender:</label>

                <div class="options">
                    <input type="radio" id="male"
                           name="gender" value="Male">
                    <label for="male">Male</label>

                    <input type="radio" id="female"
                           name="gender" value="Female">
                    <label for="female">Female</label>
                </div>

                <!-- Favorite Hobbies -->
                <label>Favorite Hobbies:</label>

                <div class="options">
                    <input type="checkbox" id="music"
                           name="hobbies" value="Music">
                    <label for="music">Listening to Music</label>

                    <input type="checkbox" id="reading"
                           name="hobbies" value="Reading">
                    <label for="reading">Reading</label>

```

```
                    <input type="checkbox" id="sports"
                           name="hobbies" value="Sports">
                    <label for="sports">Sports</label>

                    <input type="checkbox" id="travelling"
                           name="hobbies" value="Travelling">
                    <label for="travelling">Travelling</label>
                </div>

                <!-- Submit Button -->
                <button type="submit">Submit</button>

            </form>
        </section>

    </main>

    <!-- Footer -->
    <footer>
        <p>&copy; 2026 My Personal Website. All Rights Reserved.</p>
    </footer>

</body>
</html>
```

Style.css

```
/* General Styling */
* {
    margin: 0;
    padding: 0;
    box-sizing: border-box;
}

body {
    font-family: Arial, sans-serif;
    background-color: #f4f1ed;
    color: #333333;
    line-height: 1.6;
}

/* Header */
header {
    background-color: #574638;
    color: white;
    text-align: center;
    padding: 40px 20px;
}

header h1 {
    font-size: 32px;
    text-decoration: underline;
}

/* Navigation Bar */
nav {
    background-color: #8b735f;
    padding: 15px;
}

nav ul {
    list-style-type: none;
    display: flex;
    justify-content: center;
    flex-wrap: wrap;
    gap: 25px;
}
```

```
}

nav ul li a {
    color: white;
    text-decoration: none;
    font-weight: bold;
    padding: 8px 12px;
}

/* Link States */
nav ul li a:hover {
    color: #ffd699;
    text-decoration: underline;
}

nav ul li a:visited {
    color: #e5d6c8;
}

nav ul li a:active {
    color: yellow;
}

/* Main Content */
main {
    max-width: 1000px;
    margin: 30px auto;
    padding: 20px;
}

/* Sections */
section {
    background-color: white;
    padding: 25px;
    margin-bottom: 25px;
    border-radius: 10px;
    box-shadow: 0 3px 8px rgba(0, 0, 0, 0.1);
}

section h2 {
```

```
section h2 {
    color: #76563d;
    margin-bottom: 15px;
    text-decoration: underline;
}

article h3 {
    color: #8b735f;
    margin-bottom: 10px;
}

/* Hobbies List */
section ul {
    margin-left: 25px;
}

/* Video */
figure {
    text-align: center;
}

video {
    max-width: 100%;
    border-radius: 10px;
}

figcaption {
    margin-top: 10px;
    font-style: italic;
    color: #777777;
}

/* Contact Form */
form {
    display: flex;
    flex-direction: column;
    gap: 10px;
}

form > label {
```

```
    font-weight: bold;
    margin-top: 10px;
}

input[type="text"],
input[type="email"] {
    width: 100%;
    padding: 12px;
    border: 1px solid #cccccc;
    border-radius: 5px;
}

.options {
    display: flex;
    align-items: center;
    flex-wrap: wrap;
    gap: 8px;
}

.options label {
    margin-right: 15px;
}

input[type="radio"],
input[type="checkbox"] {
    cursor: pointer;
}

/* Submit Button */
button {
    background-color: #76563d;
    color: white;
    padding: 12px;
    border: none;
    border-radius: 5px;
    font-size: 16px;
    cursor: pointer;
    margin-top: 15px;
}

button:hover {
    background-color: #a17c5c;
}

/* Footer */
footer {
    background-color: #574638;
    color: white;
    text-align: center;
    padding: 20px;
}

/* Responsive Design */
@media screen and (max-width: 600px) {

    header h1 {
        font-size: 24px;
    }

    nav ul {
        flex-direction: column;
        align-items: center;
        gap: 10px;
    }

    main {
        padding: 10px;
        margin: 15px auto;
    }

    section {
        padding: 15px;
    }

    .options {
        gap: 5px;
    }
}
```

Output

Welcome to My Website

Discover more about My Personal Profile

Home About Me My Video Hobbies Contact

About Me

Introduction

Hello! My name is Fasya. I am a student who enjoys learning new things and exploring different experiences. This website is created to introduce myself, share my interests and hobbies, and provide a little information about me.

My Favorite Video



Figure 1: A beautiful flower video.

My Hobbies

Everyone has their own interests and hobbies. Here are some of my favorite activities:

- Listening to music
- Watching kdrama
- Cafe Hunting
- Travelling

Contact Me

Name:

Enter your name

Email:

Enter your email

Gender:

○ Male ○ Female

Favorite Hobbies:

☐ Listening to Music ☐ Reading ☐ Sports ☐ Travelling

Submit

© 2026 My Personal Website. All Rights Reserved.

Rubric

Table A: Rubric for Additional Lab Exercise 4 – Problem A

Criteria	Excellent (5)	Good (4)	Average (2–3)	Needs Improvement (0–1)
HTML5 Structure	All semantic tags (header, nav, section, article, footer, address, meter) are correctly used. Structure is clear and logical.	Semantic tags are mostly used correctly. Structure is good.	Some semantic tags used incorrectly or missing.	Minimal or no use of semantic tags.
Navigation Bar	Built with and styled in external CSS; no bullets or underline; hover effect works; all links jump to the correct sections.	Functional, but design could be improved.	Navigation bar present but not fully functional (e.g. some links broken).	Navigation bar missing or poorly implemented.
Video Embedding	Video correctly embedded with controls, correct source type, fallback text and a caption; scales on small screens.	Video embedded and functional, minor styling issues.	Video embedded but lacks caption, controls or proper styling.	No video or improperly embedded.
Form Design & Function	Form is complete (name, e-mail, radio, checkboxes, submit), well-structured and styled with a class.	Form is functional with minor styling issues.	Form is present but missing some elements.	Form is incomplete or missing.
CSS Styling	Effective use of internal and external CSS, at least 2 ID and 2 class selectors and 6 properties. Creative and consistent styling.	CSS applied but lacks creativity or consistency.	Some CSS applied but minimal effort shown.	Little or no CSS applied.
Text Decorations	Proper and creative use of overline, underline and linethrough (old price) across headings and links.	text-decoration used correctly but without creativity.	Minimal use of textdecoration.	No use of textdecoration.

Interactive Link States	All four link states styled in the correct order and fully functional.	Link states are functional but need more styling.	Link states applied with minimal effort or wrong order.	No link states applied.

Criteria	Excellent (5)	Good (4)	Average (2–3)	Needs Improvement (0–1)
Responsiveness & Design	Page fits small screens (viewport meta, % widths, responsive video) and is visually appealing.	Page is mostly responsive with minor design issues.	Limited responsiveness, design is basic.	Not responsive, poor design.
Total				**40 marks**

Table B: Rubric for Additional Lab Exercise 4 – Problem B

Criteria	Excellent (5)	Good (4)	Average (2–3)	Needs Improvement (0–1)
Error Identification	Identifies at least 8 errors correctly with line numbers and clear explanations.	Identifies 6–7 errors correctly.	Identifies 3–5 errors, or explanations are unclear.	Identifies fewer than 3 errors.
Error Correction & Validation	All errors corrected; page works as described and passes both the HTML and CSS validators.	Most errors corrected with 1–2 remaining validation errors.	Some errors corrected but the page still does not work correctly.	Few or no errors corrected.
External CSS & Navigation Bar	cafe.css correctly linked; navigation bar fully styled with hover effect and working links.	External CSS and navigation bar work with minor styling issues.	External CSS attempted but navigation bar is basic or partly broken.	No external CSS or navigation bar not styled.
Text Decorations & Link States	Old prices use linethrough; heading decorations and all four link states work in the correct order.	Text decoration and link states work with minor issues.	Text decoration or link states partly applied.	No text decoration or link states.
Video & Responsiveness	Video plays with controls and caption (figure/figcaption); viewport meta and responsive video applied.	Video and caption work with minor responsive issues.	Video plays but caption or responsiveness missing.	Video still not working.
Form Design & Function	Feedback form complete (name, e-mail, radio, at least 3 checkboxes, submit), styled with .feedbackform and linked in nav.	Form is functional with minor styling issues.	Form is present but missing some elements.	Form is incomplete or missing.
Total				**30 marks**